

Ensar at GroundLM 2026 Shared Tasks: Symbolic Scene Serialisation for View-Level Evidence Identification

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Abstract

An autonomous vehicle has the privilege of abundant data drawn from many sources, but abundance raises a question of provenance: when we care about one specific event in the scene, which source actually carries the evidence for it? Identifying that source matters beyond attribution, because a system that weighs every channel at once lets the irrelevant ones dilute the decision, and more input is not the same as more information. GoldenViewVQA poses this question in a controlled form, asking a system to name which of six synchronised vehicle cameras carries the evidence for a question, and to answer that question. We introduce a novel two-stage system built on symbolic scene serialisation. We sweep all six views with a detector fine-tuned on nuScenes, back-project every detection onto the ground plane, associate detections across cameras into single objects that carry an explicit view-set, and render the scene as text. A language model then names the supporting view and the answer, reading either the serialisation alone (System 1) or the serialisation together with the six raw images (System 2). The pixels help the system locate evidence, not answer questions: System 2 lifts view macro-accuracy to 0.68, against 0.38 for System 1 and 0.14 for an always-CAM_FRONT baseline. Our best run reaches 0.78 joint accuracy, 0.83 view accuracy and 0.65 view macro-accuracy on the official test split. Answer accuracy already reaches 0.95 with no image at all: option phrasing alone largely solves the answer head, leaving nearly all the measurable signal in the view head.

1 Introduction

A vehicle that drives itself is surrounded by sensors, and at any instant almost none of what they report bears on the decision it is about to make. A surround camera rig returns six simultaneous

views; the object that should trigger a brake usually appears in one or two of them. A system that consumes all six channels at once can therefore be right without being right for a reason anyone can check, because nothing in its output states which part of the scene produced the decision.

That gap is what makes view-level evidence identification pressing. When a driving stack behaves unexpectedly, an engineer has to know which view carried the evidence before the failure can be reproduced or attributed to perception rather than planning. A validation regime needs the same thing to certify that a decision followed from a real observation and not a spurious correlation. And a model that cannot isolate the relevant view has to weigh every channel equally, so redundant input dilutes a decision it ought to sharpen. Attribution is usually recovered after the fact from attention maps or saliency heuristics, which are explanations of a model rather than measurements of the scene, and which cannot be scored.

GoldenViewVQA (Wang et al., 2026a) makes the requirement measurable instead. It is one of two GroundLM 2026 shared tasks (Wang et al., 2026b); the other, LitTraceQA (Liu et al., 2026), poses the analogous grounding problem over scientific literature. We enter GoldenViewVQA only. Given six synchronised nuScenes (Caesar et al., 2020) camera images and a four-option multiple-choice question, a system must name the supporting camera view — one of the six channels or NONE_OF_THE_ABOVE — and answer the question. Submissions are ranked by joint accuracy, so credit requires both: a system cannot score by answering well while staying vague about where it looked.

Our approach is deliberately not an end-to-end model. Serialising an object-level scene representation for a language model, rather than feeding it pixels, follows Chen et al. (2024), who fuse an object-level vector modality with an LLM for explain-

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able driving, and Mao et al. (2023), who cast driving decisions as language modelling over a textual scene. Driving question answering has likewise been framed as structured reasoning over scene graphs (Sima et al., 2024). What distinguishes GoldenViewVQA from these is that the label is the *evidence source* rather than the decision, which makes per-object camera provenance — not just object presence — the quantity a serialisation must preserve.

We therefore detect agents across all six views and associate them in bird’s-eye view following Gaafar et al. (2025), and serialise the result so that each object carries the set of cameras it was seen in. The reasoning is that if every detected object is attributed to the specific cameras it appears in, then view selection reduces to grounding the question to an object and reading off its view-set. We test that reduction directly, and then test what raw pixels add on top of it.

Our contributions are:

- A symbolic serialisation of a surround-view scene that carries explicit per-object view-sets, cross-camera identity, and negative evidence, together with two systems built on it.
- Measured evidence that the pixels help view attribution significantly and do not help answering at all.
- An error analysis of the perception-to-text interface, including a failure mode where a missed detection is serialised as confirmed negative evidence.

2 Data and Preprocessing

What a record looks like. A GoldenViewVQA record is deliberately spare. It offers a question and four options, a nuScenes `sample_token` naming the instant, and a `views` map pointing at six image paths — one per camera — and nothing else. The task ships these as `eval.jsonl` (55 labelled records) alongside `eval_inputs.jsonl` and `test_inputs.jsonl` (59 records with labels withheld), together with machine-readable schemas and the official evaluator. Every question also declares a `question_group` — *causality*, *counterfactual* or *intent_prediction* — and the two splits do not draw on these evenly: development leans causality (27/55), while test spreads wider and leans intent prediction (20/59). The splits are there-

fore not distributionally interchangeable, and Section 6 shows that this is not a cosmetic difference.

Splits and label balance. Two further properties of the release shape any reasonable approach. First, the 55 labelled development records are the only labelled data on offer, so this is an inference-time task, not a training task. Second, the view labels are severely imbalanced: in development, CAM_FRONT accounts for 38/55 records while two classes hold a single example each. Because view macro-accuracy averages recall over the seven classes present in gold, each class carries roughly 14.3 points, and a single flipped prediction on a singleton class moves the headline metric by that much.

Images. The pixels arrive separately. Records point into nuScenes rather than carrying images, so we obtained `v1.0-trainval` camera data under the nuScenes Terms of Use and redistribute none of it. The 114 records turn out to reference only 438 unique frames across 66 scenes, and we confirmed all 438/438 resolve locally before running anything, since a frame that goes missing mid-run costs a full pass. Only keyframe (`samples/`) camera data is needed — no sweeps, lidar or radar — which cuts the download to a fraction of the full release.

No ground-truth annotations at inference. The task permits inferring from the images but not reading nuScenes annotations. We therefore removed every annotation-derived component from the GeoDrive pipeline (Gaafar et al., 2025): 3D bounding boxes, ego pose, the OpenStreetMap and GPS map lookups, and ground-truth future trajectories. The only nuScenes-derived quantity we retain is the camera rig geometry, which we treat as published hardware specification: across all 850 trainval logs, focal length varies by $\leq 1.3\%$ and mount positions by under 6 cm, so we bake per-camera intrinsics and extrinsics into constants. Inference reads six JPEG files and touches no nuScenes table.

3 System Architecture

3.1 Shared perception front-end

The front-end converts six images into a list of objects, each annotated with the set of cameras it was detected in.

Detection. We run YOLOv8-m (Jocher et al., 2023), fine-tuned on nuScenes over seven agent

classes (car, truck, bus, trailer, motorcycle, bicycle, pedestrian) as in GeoDrive (Gaafar et al., 2025), independently over each of the six images. We set the confidence threshold to 0.15, deliberately low: a missed object is unrecoverable in the serialised text, whereas a false positive can still be discounted downstream by a model that sees pixels.

Ground-plane projection. For each detection we back-project the bottom-centre of the box through the fixed rig geometry onto the plane $z = 0$ in the ego frame, giving a metric position. Rays that do not meet the plane in front of the camera are discarded.

Cross-camera association. Detections of the same class within 2.0 m in bird’s-eye view — one vehicle width, following Gaafar et al. (2025) — are merged into a single object. Critically, and unlike the GeoDrive implementation, a merged duplicate is *absorbed* rather than discarded, so the camera it came from is recorded. The resulting `views` list is the quantity the task actually asks about; discarding it makes the full rendering below impossible.

3.2 Serialisation

Objects are rendered as text in two forms. The **full render** emits one section per camera in fixed order, objects sorted near-to-far, with one shared identifier per object across every section it appears in, explicit cross-references (`[same object also in CAM_FRONT]`) generated from the view-set, and an explicit `no agents detected` line for empty cameras. The **isolated render** emits a single camera with cross-references stripped and identifiers local to that view.

Bearing is reported in degrees, since azimuth is the well-conditioned axis of monocular geometry and is what view attribution depends on. Range is quantised into bands (near < 10 m, mid 10–25 m, far > 25 m) rather than reported as a point estimate, because flat-ground monocular range error grows quadratically with distance.

3.3 Submitted systems

System 1 (text only). The full render, the question and the four options are given to a language model, which returns the predicted view and answer. No images are passed. This isolates what the symbolic layer contributes.

System 2 (text and pixels). The full render is presented first, followed by all six images, each

labelled with its camera name. The prompt states that the detector cannot see traffic lights, signs, cones, road markings, lane geometry or barriers, and instructs the model to trust the pixels where text and pixels disagree.

Prompts. The per-record prompt is generated by our sample builders from the serialised scene, the question and the options. Separate runner instructions govern how records are worked through, how text is weighed against pixels, and the output contract; these sit verbatim in the `prompts/` directory, one file per system.

Models and compute. We ran the serialisation through two hosted reasoning models: Claude Opus 5 with extended thinking, and GPT-5.6 Sol at its highest reasoning setting. Within each model the prompt is held fixed across systems, so the System 2 versus System 1 contrast measures the contribution of pixels rather than a change of model. Our best official submission used Claude Opus 5. Detection ran on a single consumer GPU in under two minutes for all 438 images. The 114 records were processed in shards of 9–11 records to bound context per call.

System 3 (per-view evidence ranking). We additionally implemented an approximation of an information-gain formulation. Each camera is presented on its own, with the isolated render and that camera’s image, and the model reports an answer and an ordinal evidence-strength score for that view; a question-only pass supplies a prior answer. The predicted view is the argmax of evidence strength. This is an approximation and we label it as such: an ordinal judgment elicited in natural language is a coarse substitute for the distributional divergence the formulation calls for. It costs seven passes per question instead of one.

4 Experimental Setup

We report the four official metrics: `view_accuracy` (micro), `view_macro_accuracy` (mean per-view recall over classes present in gold), `answer_accuracy`, and `joint_accuracy`. All numbers come from the organizers’ `evaluate.py`, imported rather than reimplemented, so our reported metrics cannot drift from the official definition.

System	View	Macro	Ans.	Joint
<i>label-only baselines</i>				
uniform_random	0.146	0.223	0.327	0.091
prior_random	0.491	0.102	0.309	0.200
constant	0.691	0.143	0.255	0.182
majority	0.691	0.143	0.309	0.200
<i>Claude Opus 5</i>				
System 1 (text)	0.709	0.382	0.945	0.709
System 2 (+pixels)	0.873	0.679	0.927	0.855
<i>GPT-5.6 Sol</i>				
System 1 (text)	0.727	0.481	0.909	0.691
System 2 (+pixels)	0.909	0.812	0.945	0.891
System 3 (per-view)	0.873	0.746	0.909	0.836

Table 1: Development-set results ($n = 55$), official evaluator.

Metric	Δ	95% CI
view_accuracy	+0.164	[+0.073, +0.273]
view_macro_accuracy	+0.297	[+0.078, +0.461]
answer_accuracy	-0.018	[-0.055, +0.000]
joint_accuracy	+0.146	[+0.036, +0.255]

Table 2: Paired bootstrap, System 2 minus System 1, 4000 resamples.

Baselines. We include four label-only baselines that read neither text nor pixels: *constant* (always CAM_FRONT, answer A), *majority* (most frequent gold view and answer), *prior_random* (sampled from the development label distribution), and *uniform_random*.

Uncertainty. With $n = 55$, point estimates are close to meaningless, so we report 95% percentile bootstrap intervals over questions (2000 resamples). For system comparisons we use a *paired* bootstrap that resamples the same questions for both systems, so per-question difficulty cancels. To calibrate how much of an observed difference is noise, we ran *uniform_random* across 20 seeds, changing nothing else: macro accuracy had mean 0.112, standard deviation 0.069, and range [0.015, 0.223]. A 0.208-wide spread arises from the random seed alone, which is wider than the gap between several of our baselines.

5 Results

Table 1 gives development results. System 2 improves view macro-accuracy from 0.143 (constant baseline) to 0.679, and joint accuracy from 0.182 to 0.855.

Table 2 isolates the contribution of the pixels. The view metrics improve with intervals excluding zero. Answer accuracy does *not* improve; its point

Submitted run (System 2)	View	Macro	Ans.	Joint
Constant baseline	0.780	0.167	0.237	0.203
GPT-5.6 Sol	0.780	0.562	0.932	0.763
Claude Opus 5	0.831	0.652	0.915	0.780

Table 3: Official test results ($n = 59$) from the GroundLM 2026 evaluator.

Gold view	n	Sys. 1	Sys. 2
CAM_FRONT	38	0.921	1.000
CAM_FRONT_LEFT	8	0.250	0.750
CAM_FRONT_RIGHT	2	0.500	1.000
CAM_BACK	1	1.000	1.000
CAM_BACK_LEFT	2	0.000	0.000
CAM_BACK_RIGHT	1	0.000	1.000
NONE_OF_THE_ABOVE	3	0.000	0.000

Table 4: Per-class view recall on development.

estimate is negative and its interval has upper bound exactly 0.000.

The answer head is nearly saturated without vision. System 1 reaches 0.9455 answer accuracy having seen no image. Taken with Table 2, this indicates the four options are separable from text alone, and that the remaining headroom is essentially all in view attribution. Note that System 1’s joint accuracy exactly equals its view accuracy: answer accuracy is high enough that joint is determined by the view head.

Official test results. Table 3 reports every GoldenViewVQA run we submitted to the evaluator Space. Our best submission reaches 0.7797 joint accuracy with 0.8305 view accuracy and 0.6522 view macro-accuracy.

6 Error Analysis

Evidence-selection failures. Table 4 localises the remaining error. System 2 is perfect on four of seven classes. All residual loss is concentrated in CAM_BACK_LEFT (0/2) and NONE_OF_THE_ABOVE (0/3), together 28.6 macro points. Both systems predicted NONE_OF_THE_ABOVE zero times in every run, despite prompts naming it as a valid and rare label. Both CAM_BACK_LEFT errors went to CAM_FRONT_LEFT: a left-side front/rear confusion, not a left/right confusion.

Attention collapse. System 1 predicted CAM_FRONT on 44/55 development records against a gold count of 38, and lost 6 of 8 CAM_FRONT_LEFT cases to it. Adding pixels

reduced this collapse substantially, which is the main mechanism behind the macro gain.

The ceiling is representational, not perceptual.

We went looking for missed detections and did not find them. At threshold 0.15 the front-end yields 29.3 raw detections and 22.0 associated objects per frame on development, against roughly 27 annotated agents per frame in nuScenes, and no frame came out empty. The detector is not failing to see; it has no vocabulary for what the questions are about. Decisive evidence in our records was repeatedly a lowered car-park gate arm, a stop sign, a traffic signal state, a splitter island, cones, chevron boards, a painted lane arrow or a crosswalk — not one of which a seven-class agent detector can name. No amount of extra recall reaches these objects, which is why System 1 plateaus so far below System 2.

Serialisation faults. Inspecting the pixel-versus-text disagreements our System 2 runs reported surfaced three distinct faults at the perception-to-text interface, in descending order of severity.

(i) *False negative rendered as confirmed negative.* Several cameras were serialised as `no agents detected` while the image contained vehicles, in one case a truck bed filling the entire frame at close range. This is the most damaging fault available in our design, because the full render presents an empty camera as grounds to rule that view out. We suspect boxes whose bottom edge reaches the image border are dropped by the ground-plane projection; clamping and flagging the object as *very near*, *range uncertain* would be preferable to silently emitting a negative.

(ii) *View-set misattribution.* In at least one case, objects fully visible in `CAM_FRONT` were attributed only to `CAM_FRONT_LEFT`, where they were clipped at the frame edge. The fixed 2.0 m association radius is likely too tight at range, where flat-ground projection error grows quadratically; a range-dependent radius is the natural fix. This fault matters more than a false positive because view attribution is precisely what is scored.

(iii) *Phantom objects.* Cars parked in multi-storey car parks, vehicles on the far side of a kerbed median, and a person seated on a grass embankment were all projected into the scene as road agents. Under System 2 the pixels arbitrate; under System 1 there is no confidence signal a text-only reader could use to discount them.

Irreducible ambiguity. A recurring pattern is that the decisive object is clipped at the frame edge of one camera while fully visible in another, so more than one view is defensible while gold names one. We expect this to be a floor on achievable view accuracy rather than a fixable deficiency.

Development versus test. Official test performance is below development, as anticipated, but the size of the gap differs sharply between our two submitted runs and that difference is the more interesting result. The Claude Opus 5 run moved from 0.679 macro on development to 0.652 on test, a drop of 0.027 that sits far inside the development interval. The GPT-5.6 Sol run moved from 0.812 to 0.562, a drop of 0.250. The run with the stronger development score generalised markedly worse, which is what selecting on a 55-record development set will do: the apparent development advantage was substantially split-specific. Under the official metric the ordering of our two runs reverses between splits.

Test is also harder in composition. It leans more on *intent_prediction* (20/59 versus 13/55), our weakest group on development (0.692 view accuracy, against 0.926 on causality).

7 Conclusion

Serialising a surround-view scene into text with explicit per-object view-sets turns view-level evidence identification into a grounding-and-lookup problem and takes view macro-accuracy from 0.143 to 0.382 with no visual input. Supplying the raw images on top of that serialisation nearly doubles it again to 0.679, and a paired bootstrap confirms the gain is in the view head only.

The most useful thing we can report is negative. Answer accuracy of 0.9455 without any image indicates that the answer options are largely separable from text alone, and that joint accuracy therefore behaves as a view metric. Future work on this benchmark should concentrate on view attribution.

A second lesson is methodological. Our two models' rankings reverse between splits: the run that scored 0.812 macro on development fell to 0.562 on test, while the weaker development run (0.679) held at 0.652. On a 55-record development set, selecting a submission by development score is close to selecting noise.

For our own system, the clearest next steps follow from Section 6: never serialise an unverified negative, make the association radius range-

dependent, and add an explicit abstention rule so that `NONE_OF_THE_ABOVE` becomes reachable rather than structurally impossible.

Limitations

Our evaluation rests on 55 labelled records. Bootstrap intervals are correspondingly wide — 0.306 for System 2 macro accuracy — and two view classes contain a single example each, so per-class recalls for those classes are one-bit measurements. We showed that changing only a random seed moves macro accuracy over a 0.208 range; readers should not treat differences smaller than that as meaningful.

The rig geometry we hard-code is nuScenes-specific and would not transfer to another sensor platform without re-derivation. The perception front-end is monocular and assumes flat ground, which is the direct cause of the phantom-object and misattribution faults documented above.

Our systems consume exactly the six images of the annotated keyframe and nothing else. Because each record identifies a nuScenes sample within a continuous scene, temporally adjacent frames are available, and a similarity-based retrieval step — gathering nearby frames whose scene content is close to the queried keyframe and serialising them jointly — would give substantially richer context than a single instant. This would particularly help the *intent_prediction* group, our weakest at 0.692 view accuracy, where a pedestrian’s direction of travel or a vehicle’s manoeuvre is far more legible over a short window than in one frame. It would also supply the motion cues our single-frame design discards by construction.

Both submitted systems depend on a cloud-hosted multimodal model. Beyond cost and reproducibility concerns — a hosted model may be updated or withdrawn, so our exact numbers may not be reproducible later — the latency is disqualifying for the application domain this benchmark is drawn from. Each System 2 record requires transmitting six full-resolution images and waiting on a remote inference call, which is orders of magnitude away from the budget of an on-vehicle perception loop. A safety-relevant evidence-attribution component would have to run locally. Investigating smaller open-weight vision-language models, and distilling the behaviour of the larger model into one, is the more practical direction, and it is the direction our prior work took for its reasoning module

(Gaafar et al., 2025). We expect a meaningful accuracy cost, and quantifying that trade-off is exactly what makes the experiment worth running.

Ethics Statement

Images were obtained from the official nuScenes distribution under its Terms of Use. No images are redistributed with our code or in this paper, consistent with the shared task’s own policy. We use no personal data beyond what the public nuScenes release already contains; pedestrians appear in the source imagery as distributed by nuScenes, and we perform no identification of individuals.

All external resources are disclosed in Section 2 and Section 3: the nuScenes dataset, a YOLOv8-m checkpoint fine-tuned on nuScenes from our prior work, and a commercial multimodal language model accessed through its provider. No synthetic or generated training data was used.

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